Curriculum Vitae

Lingyu Zhang

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Google scholar: https://scholar.google.com/citations?user=FLZvf6MAAAAJ&hl

EDUCATION:

Rensselaer Polytechnic Institute, Troy, NY
Ph.D. in Electrical Engineering, Aug 2021, GPA: 3.93/4.00

Columbia University, New York, NY
Master of Science in Electrical Engineering, Dec 2017, GPA: 3.71/4.00

Huazhong University of Science and Technology, Wuhan, China
Bachelor of Engineering in Optoelectronic Information Engineering, June 2016, GPA: 3.83/4.00

RESEARCH TOPIC:

Multimodal Foundation Models, Self-Supervised Learning, Representation Learning, Temporal Transformers, Time-Series Foundation Models, Multimodal Machine Learning, Health AI, Agentic AI Systems, Autonomous Reasoning & Planning

WORK EXPERIENCE:

Apple, Cupertino, CA — July 2025 - Present

Generative AI Engineer - Health Sensing

Leading the development of next-generation AI systems for personalized health, spanning **Multimodal Foundation Models, Agentic AI**, and large-scale health intelligence platforms that transform physiological, behavioral, and contextual data into actionable user insights and adaptive health experiences.

- Developing large-scale **Multimodal Health Foundation Models** that jointly learn from wearable sensor streams, behavioral signals, and contextual health data to enable generalized health understanding and personalization.
- Designing **Self-Supervised Learning**, **Multimodal Representation Learning**, and **Foundation Model Pretraining** frameworks to learn transferable physiological embeddings from heterogeneous time-series data at scale.
- Building scalable **Distributed Training**, data processing, and evaluation pipelines for billions of sensor observations across diverse health, activity, and behavioral domains.
- Advancing **Temporal Transformers**, **Long-Context Sequence Modeling**, **Multimodal Fusion**, and **Transfer Learning** techniques for health forecasting, risk prediction, and downstream health sensing tasks.
- Architecting production-grade **Agentic AI Systems** with **RAG**, **Tool-Augmented LLMs**, **Adaptive Planning**, and **Contextual Memory** to deliver personalized health recommendations and insights.
- Designing automated **LLM-as-a-Judge Evaluation Frameworks** and driving **On-Device Inference Optimization** to ensure reliable, privacy-preserving, and scalable deployment of AI systems in production.

Samsung Research America, Mountain View, CA Sep 2022 - July 2025
Senior Research Scientist, Multimodal AI & Knowledge and Dialogue
A1 Inventor Award, Jan 2024

- Led research and development of **Personalized Language Models** for on-device question answering, leveraging user context, retrieval, and efficient adaptation techniques to deliver high-quality experiences under strict latency and compute constraints.
- Designed and optimized **Multimodal Language Models** integrating visual, textual, and structured knowledge for visual question answering, knowledge grounding, and reasoning-intensive conversational AI applications.
- Developed **Small Language Model (SLM)** planning and reasoning systems with tool-use and task decomposition capabilities, enabling efficient execution of multi-step workflows on resource-constrained devices.
- Advanced **Multimodal Representation Learning** and compositional zero-shot learning techniques to improve generalization across unseen concepts, domains, and user-specific contexts.
- Built customized **Diffusion Models** and personalization frameworks for context-aware image generation and visual content understanding, enhancing model adaptability and user-centric experiences.
- Collaborated with cross-functional research and product teams to translate cutting-edge multimodal AI research into scalable consumer-facing intelligent systems.

DiDi Research America, Mountain View, CA Nov 2021 - Sep 2022
Software Engineer II in Autonomous Driving - Perception team
New Hire Award, Aug 2022

- Designed and deployed **Multimodal Fusion Models** combining camera, LiDAR, and geometric features for cyclist heading estimation and motion understanding in safety-critical autonomous driving systems.
- Developed high-performance perception models for **Small Object Detection**, leading model architecture design, feature representation learning, and large-scale training optimization to improve detection accuracy and robustness.
- Built end-to-end training and evaluation pipelines for multimodal perception systems, supporting efficient experimentation, model validation, and production deployment.
- Researched and implemented state-of-the-art **Computer Vision**, **3D Perception**, and multimodal learning techniques to improve scene understanding across diverse real-world driving environments.
- Optimized model inference and deployment strategies to meet real-time latency and reliability requirements for large-scale autonomous driving applications.
- Collaborated closely with planning, and infrastructure teams to improve robustness and scalability across the autonomous driving stack.

INTERNSHIP EXPERIENCE:
American International Group (AIG), New York, NY September 2017 – December 2017
3D reconstruction for car accident damage estimation from single 2D image *Position: Research Assistant*

SELECTED PUBLICATIONS:

- S. Paul, **L. Zhang**, Y. Shen, and H. Jin. Enabling device control planning capabilities of small language models. 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (**ICASSP**).
- J. S. Smith, Y. C. Hsu, **L. Zhang**, T. Hua, Z. Kira, Y. Shen, and H. Jin (2024). Continual diffusion: Continual customization of text-to-image diffusion with c-lora. Transactions on Machine Learning Research (**TMLR**).
- **L. Zhang**, T. Hua, Y. Shen, and H. Jin (2023). Prompt Tuning for Zero-shot Compositional Learning. arXiv preprint arXiv:2312.02191.
- **L. Zhang** and R.J. Radke, Natural Language Video Moment Localization Through Query-Controlled Temporal Convolution, 2022 IEEE Winter Conference on Applications of Computer Vision (**WACV**).
- **L. Zhang** and R.J. Radke, Temporal Attention and Consistency Measuring for Video Question Answering, ACM International Conference on Multimodal Interaction (**ICMI**), October 2020.
- **L. Zhang** and R.J. Radke, A Multi-Stream Recurrent Neural Network for Social Role Detection in Multiparty Interactions. **IEEE Journal** of Selected Topics in Signal Processing (2020).
- **L. Zhang**, I. Bhattacharya, M. Morgan, M. Foley, C. Riedl, B.F. Welles, and R.J. Radke, Multiparty Visual Co-Occurrences for Estimating Personality Traits in Group Meetings, 2020 IEEE Winter Conference on Applications of Computer Vision (**WACV**). IEEE, 2020.
- **L. Zhang**, M. Morgan, I. Bhattacharya, M. Foley, J. Braasch, C. Riedl, B.F. Welles, and R.J. Radke. Improved Visual Focus of Attention Estimation and Prosodic Features for Analyzing Group Interactions. ACM International Conference on Multimodal Interaction (**ICMI**), October 2019.
- M. Li, **L. Zhang**, R.J. Radke, and H. Ji, Keep Meeting Summaries on Topic: Abstractive Multi-Modal Meeting Summarization, Annual Meeting of the Association for Computational Linguistics (**ACL**), July 2019.

TEACHING EXPERIENCE

ECSE-2500 **Engineering Probability** — January 2019 – May 2019
Rensselaer Polytechnic Institute, Troy, NY — *Position: Teaching assistant*

ECSE-4750 **Computer Graphics** — September 2018 – December 2018
Rensselaer Polytechnic Institute, Troy, NY — *Position: Teaching assistant*

E6885 **Reinforcement Learning** — September 2017 – December 2017
Columbia University, New York, NY — *Position: Course assistant*

Journal Reviewer: Journal of Computer Vision and Image Understanding, Frontiers in Big Data, Information Processing Management, Transactions on Knowledge and Data Engineering
Conference Reviewer: EMNLP, ACL, WACV